

American Samoa Model Checks

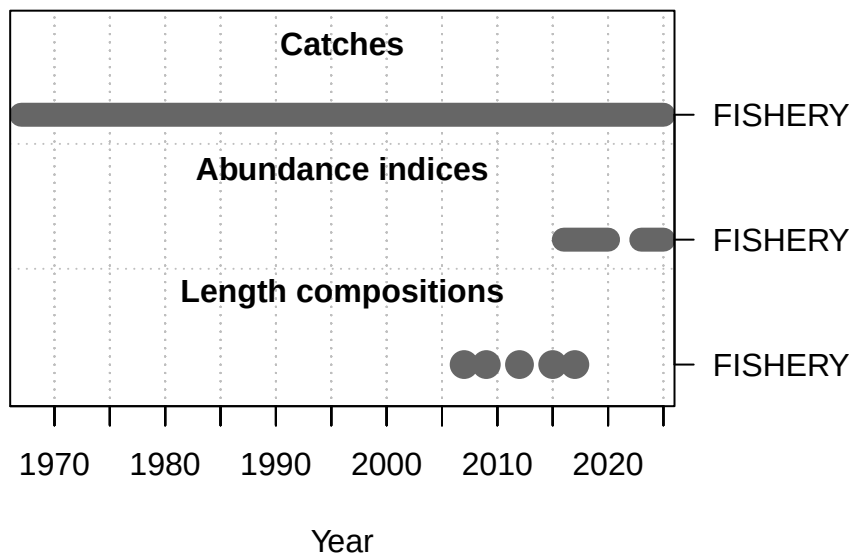
Marc Nadon and Meg Oshima

2026-04-06

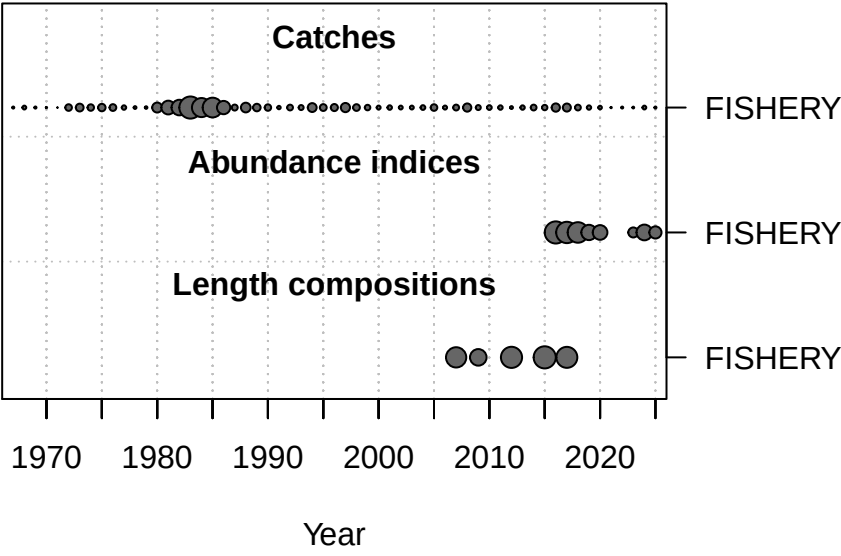
This is a summary report for the PRZO base model run.

Model Output

Input Data



Fleet	RMSE.perc	Nobs
FISHERY	98.1	8
Combined	98.1	8



Convergence Check

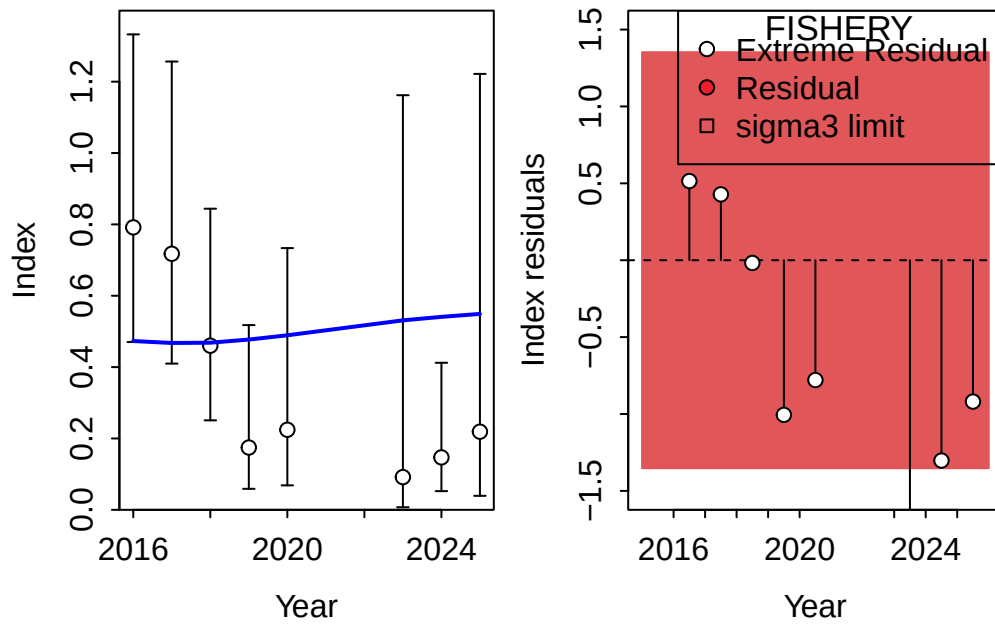
```
Converged      MaxGrad
1      TRUE 3.59117e-06
```

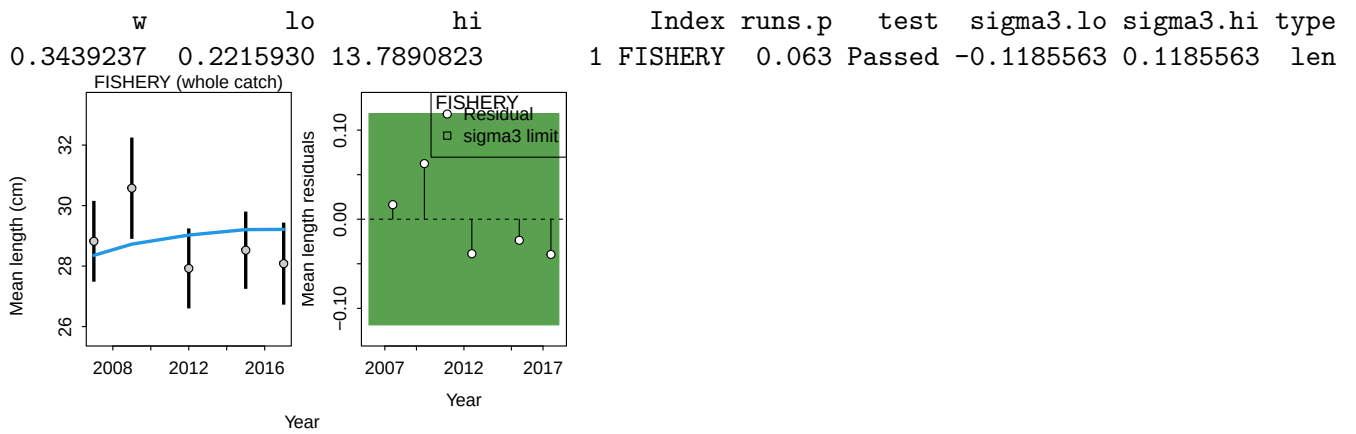
```
[1] "1 NOTE:  Max data length bin: 40  < max pop len bins: 44; so will accumulate larger pop
[2] "N warnings: 1"
```

Fleet	RMSE.perc	Nobs
FISHERY	3.9	5
Combined	3.9	5

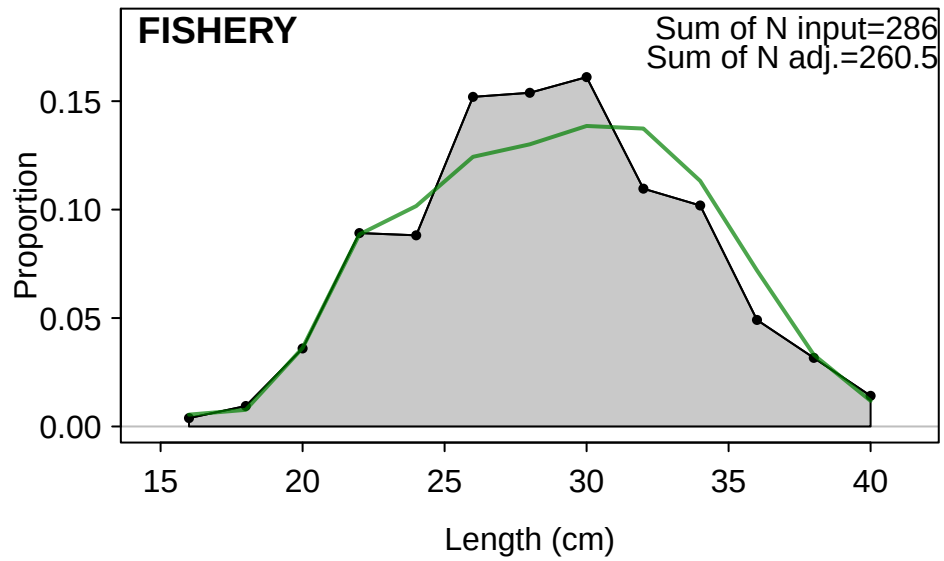
Fit to Model

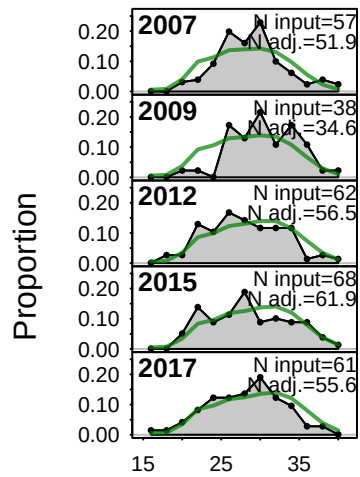
CPUE





Length Comp



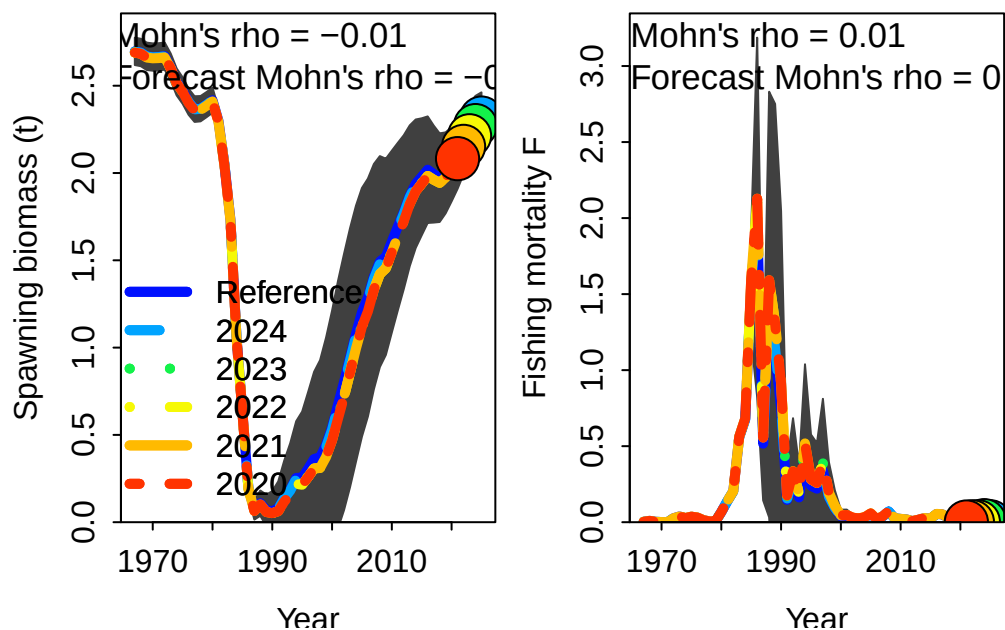


Length (cm)

Retrospective and Hindcasting

Retrospective

Mohn's Rho stats, including one step ahead forecasts:



Mohn's Rho stats, including one step ahead forecasts:

	type	peel	Rho	ForecastRho
1	F	2024	0.001498467	0.001388471
2	F	2023	0.008063130	0.007403610
3	F	2022	0.010384227	0.009434170
4	F	2021	0.011500803	0.010384227
5	F	2020	0.012775291	0.011500803
6	F Combined		0.008844383	0.008022256

Hindcasting

Plotting Hindcast Cross-Validation (one-step-ahead)

Computing MASE with only 3 of 5 prediction residuals for Index FISHERY

Warning: Unequal spacing of naive predictions residuals may influence the interpretation of

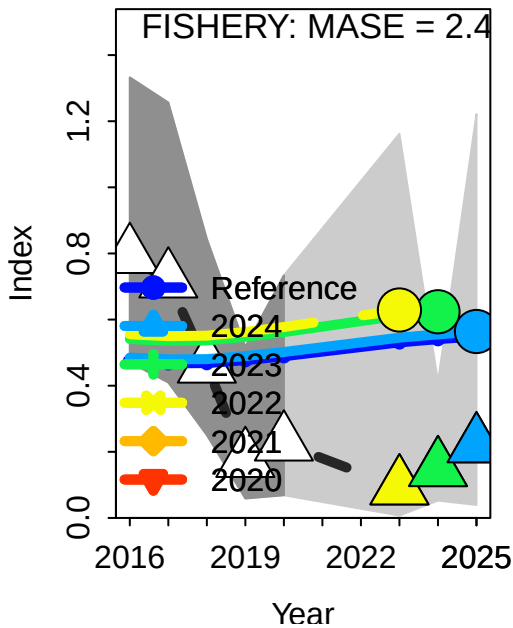
MASE stats by Index:

Plotting Hindcast Cross-Validation (one-step-ahead)

No observations in evaluation years to compute prediction residuals for Index FISHERY

MASE stats by Index:

Index	Season	MASE	MAE.PR	MAE.base	MASE.adj	n.eval
1 FISHERY	1	NA	NA	NA	NA	0



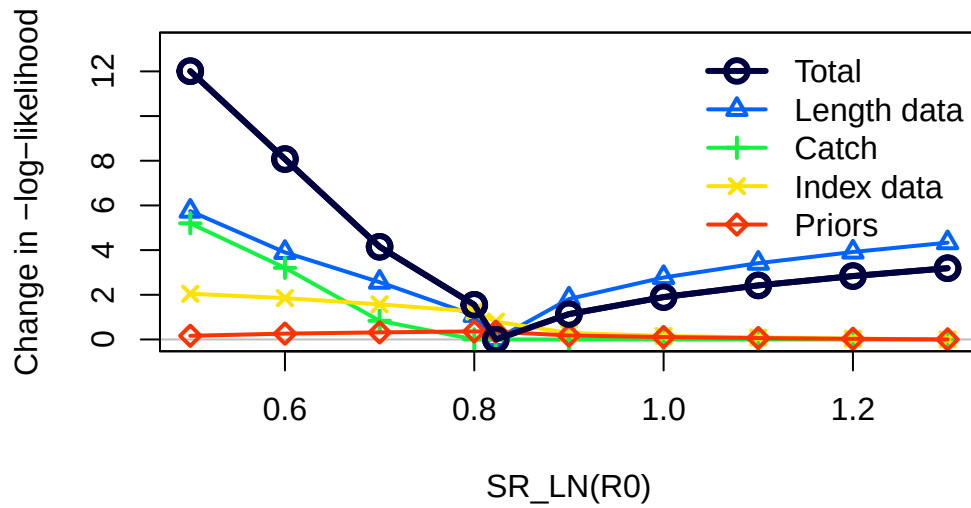
Recruitment Deviations

Likelihood Profile

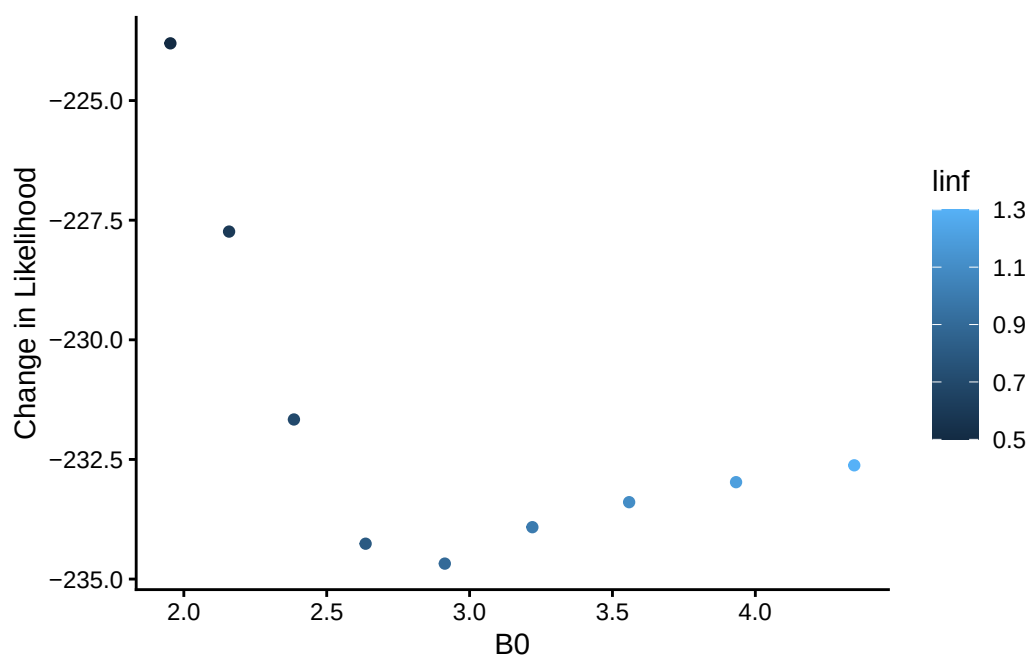
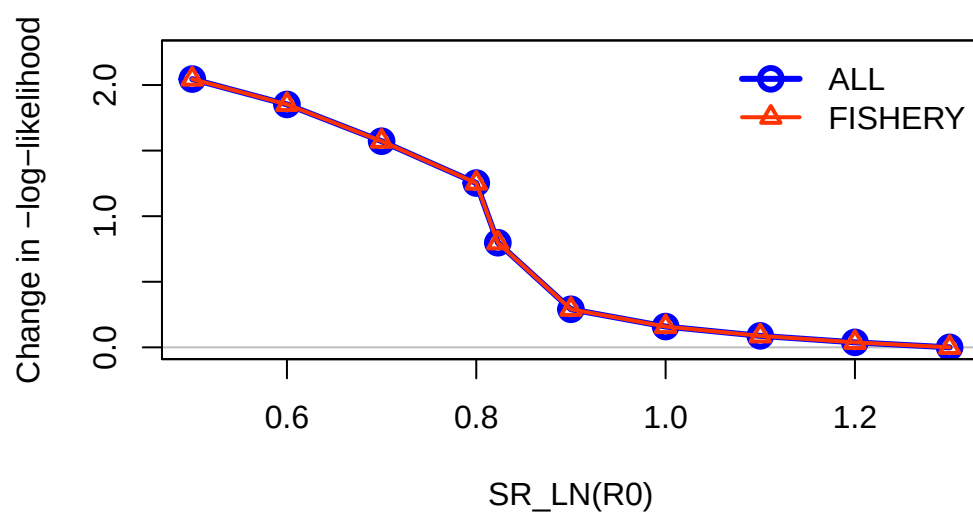
[1] "SR_LN(R0)"

	frac_change	include	label
TOTAL	1.0000	TRUE	Total
Catch	0.4329	TRUE	Catch
Equil_catch	0.0000	FALSE	Equilibrium catch
Survey	0.1703	TRUE	Index data
Length_comp	0.4784	TRUE	Length data
Recruitment	0.0000	FALSE	Recruitment

InitEQ_Regime	0.0000	FALSE	Initital equilibrium recruitment
Forecast_Recruitment	0.0000	FALSE	Forecast recruitment
Parm_priors	0.0300	TRUE	Priors
Parm_softbounds	0.0000	FALSE	Soft bounds
Parm_devs	0.0000	FALSE	Parameter deviations
Crash_Pen	0.0000	FALSE	Crash penalty

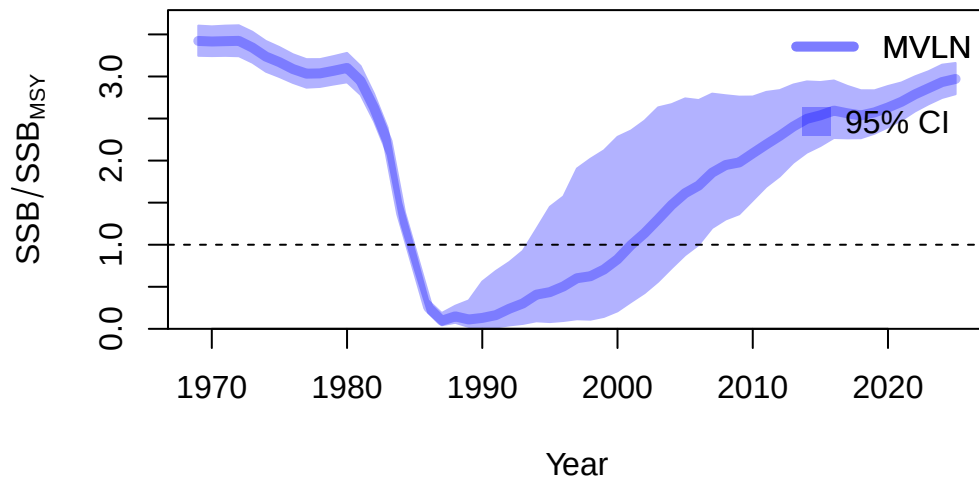
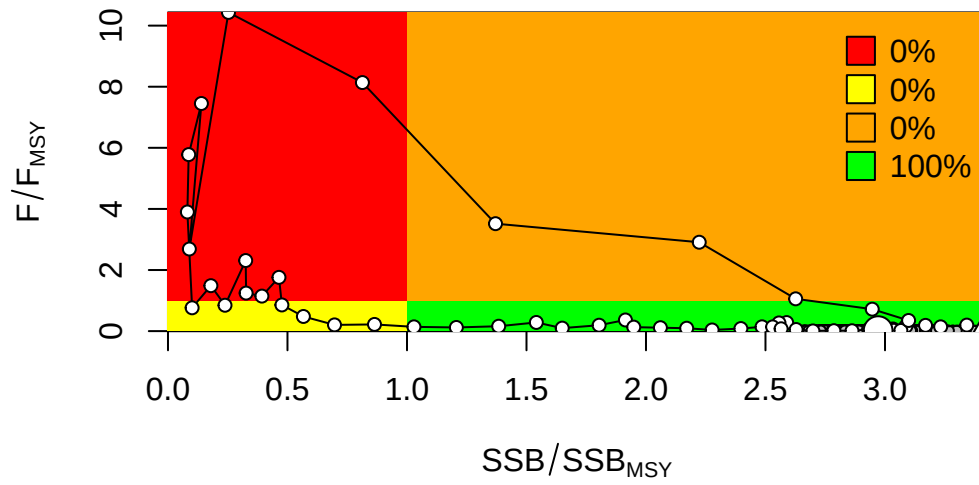


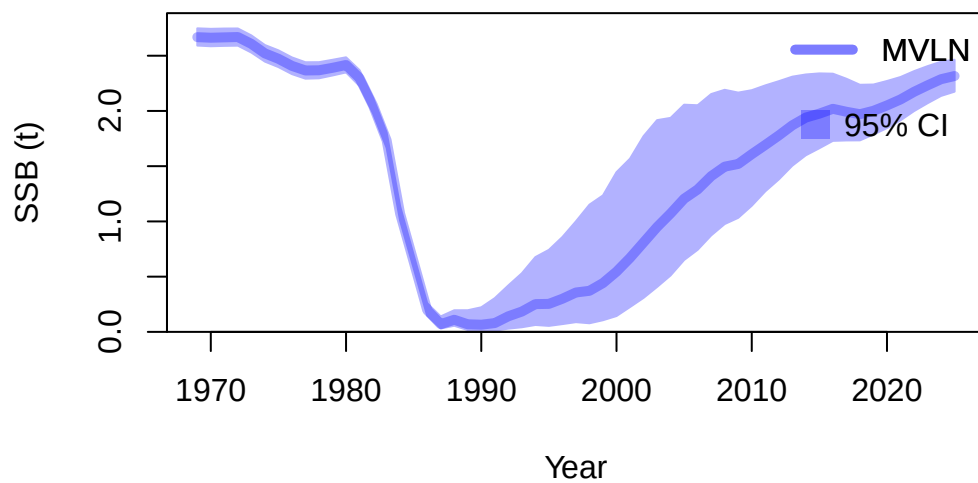
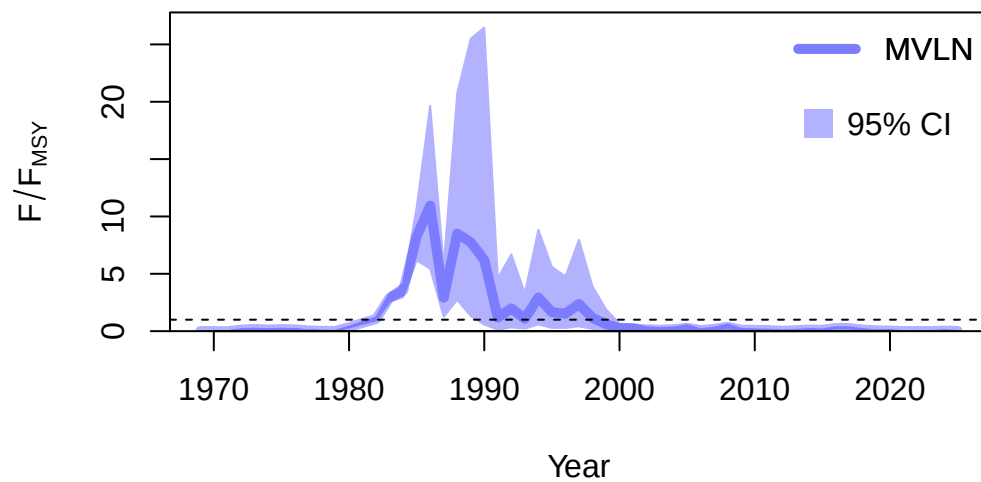
Changes in survey likelihood by fleet

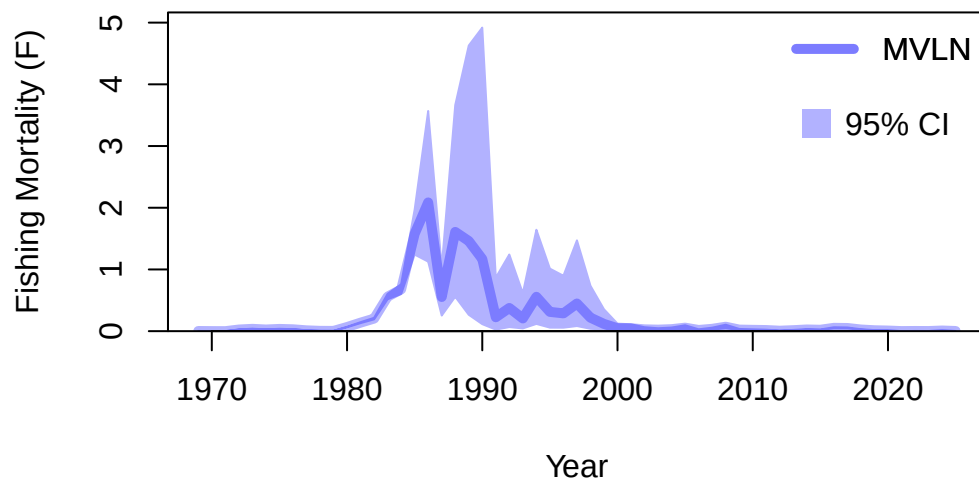


Management Quantities

starter.sso with Bratio: SSB/SSB_{MSY} and F: _abs_F

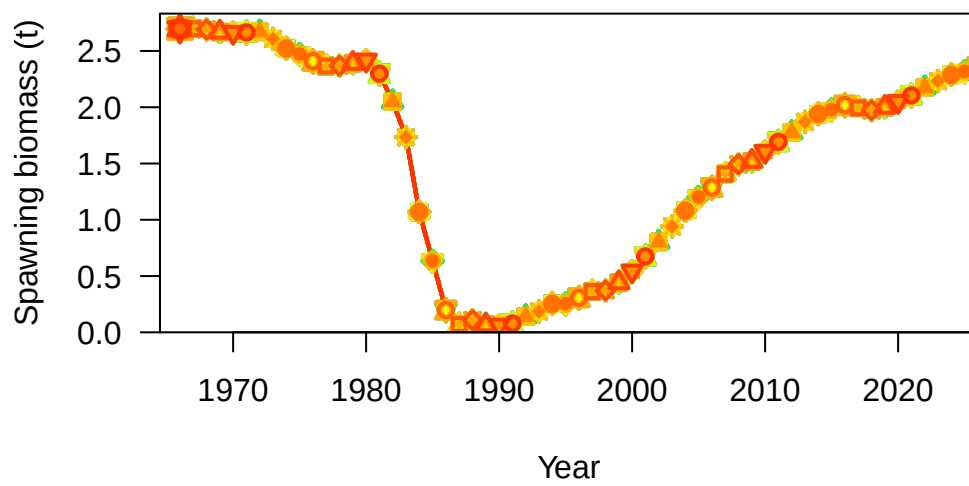
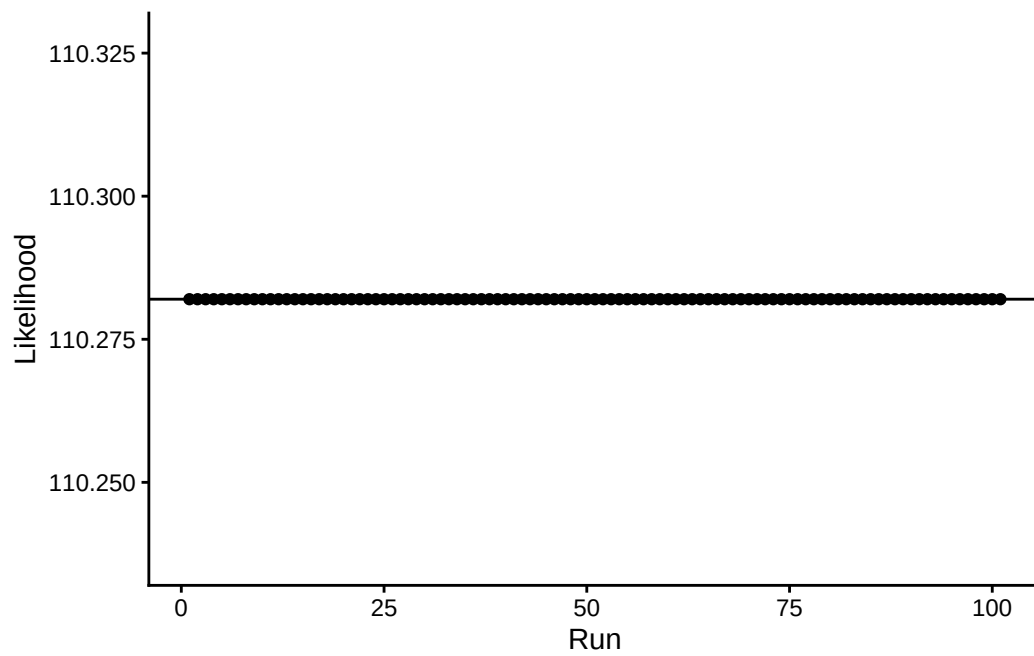


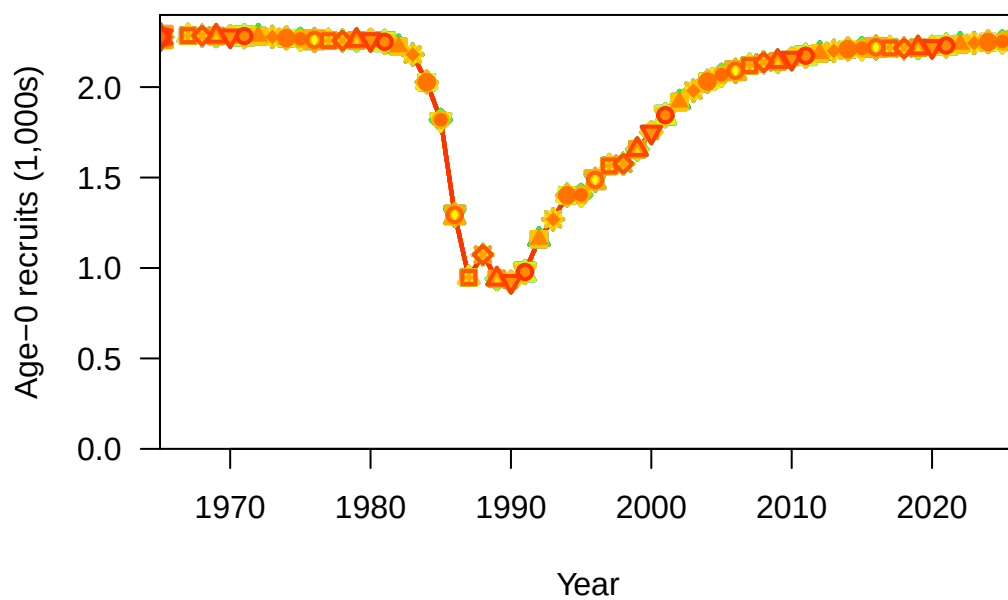
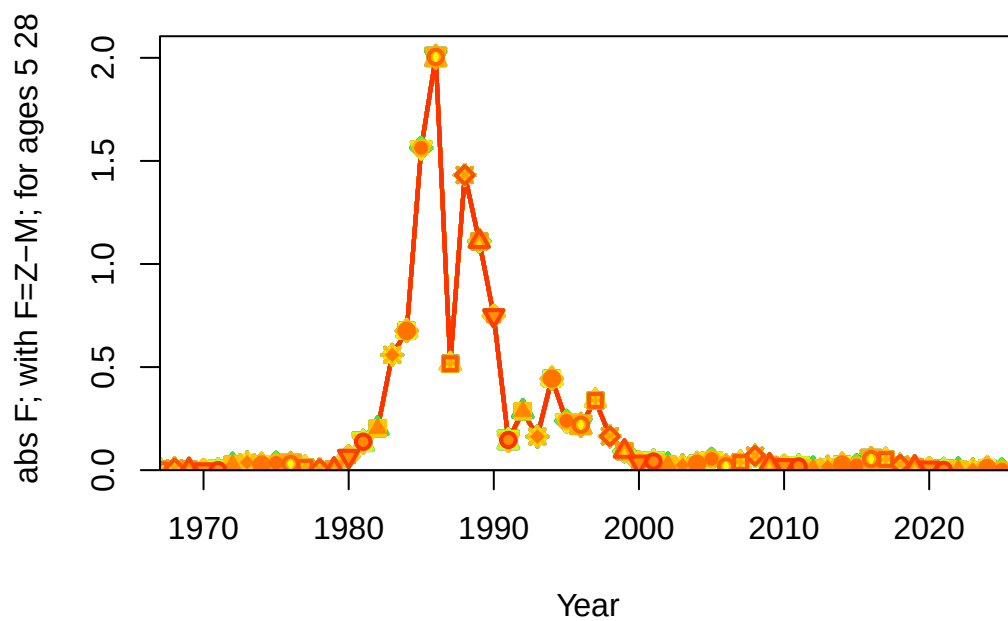




null device
1

Jitter





Selectivity and Maturity

